A Better Start: Sensitivity-Aware Warm-Up for Robust and Efficient Fine-Tuning

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Background & Motivation: The Safety Basin Perspective in LLM Fine-Tuning

Safety Basin in LLM Fine-Tuning.

During pre-training, Large Language Models (LLMs) develop a Safety Basin—a stable, reliable region in the parameter space where the model exhibits aligned and robust behavior.

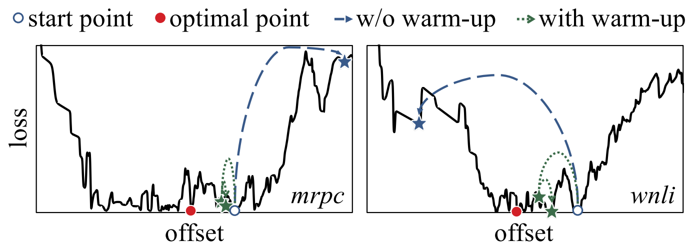
How Traditional Warm-Up Works.

Traditional warm-up acts as a safeguard to preserve this pre-trained knowledge:

- **Preventing Divergence:** By starting with a tiny learning rate, it prevents aggressive early gradients from "kicking" the model out of the Safety Basin.
- **Smoothing Transitions:** It allows the model to cautiously adapt to new downstream data while staying within the boundaries of stable, low-loss regions.

However, ...

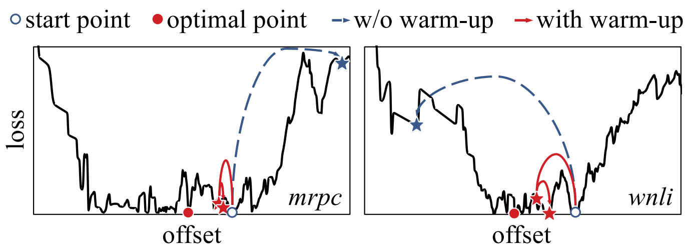
- **Landscape Blindness:** Static schedules ignore local geometry, often settling into nearby suboptimal sub-basins.
- **Hyperparameter Fragility:** Highly sensitive to warm steps and random seeds, causing inconsistent convergence.
- **Static Inefficiency:** Heuristic-based growth is often unnecessarily slow, wasting early training budget.



Methodology: Sensitivity-Aware Warm-Up (SAWU)

Core Ideas of our SAWU.

Instead of a "blind" increase, SAWU monitors the landscape's response. It dynamically leverages learning sensitivity to guide updates toward deeper and more stable basins.



Detailed Technologies in our SAWU.

- **Sensitivity Monitoring:** Actively tracks how the loss responds to learning rate changes to identify regions.

$$s_i = \frac{1}{2}(1 - \cos\theta_i) = \frac{1}{2}\left(1 - \frac{v_{i-1}\cdot v_i}{||v_{i-1}||\cdot||v_i||+\epsilon}\right)$$

$$\mathcal{L}_{SA} = \left(1 + \alpha\cdot\left(S_t - \frac{1}{2}\right)\right)\cdot\mathcal{L}_t$$

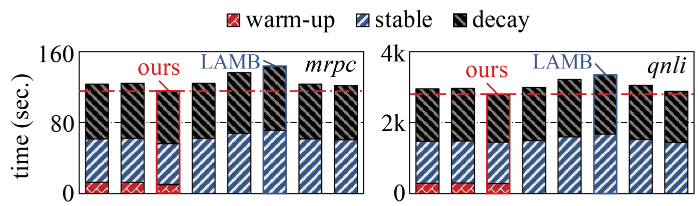
- **Adaptive Scheduling:** Automatically adjusts the warm-up pace based on the real-time geometry of the loss landscape.
- **Phase Transition Strategy:** A seamless, automated shift across Warm-up, Stable, and Decay.

Experimental Results

method	mrpc	qnli	mnli	qqp	cola	stsb
vanilla	83.71	50.54	35.45	63.18	0.00	87.75
+decay	84.06	50.54	31.82	67.21	34.56	89.19
+warm-up	85.86	90.24	84.44	85.89	55.75	89.87
AdamW	85.57	90.26	83.85	86.51	53.19	89.71
RAdam	86.55	49.46	81.15	63.18	61.12	90.01
LARS	66.49	86.66	81.84	85.49	48.02	83.26
LAMB	86.03	90.95	86.49	90.23	57.79	88.76
RV-AdamW	66.49	50.59	35.45	63.18	57.21	88.48
RV-Lion	66.49	49.46	35.45	63.18	56.12	88.12
ours	**87.59**	**92.62**	**87.00**	**91.01**	**63.12**	**90.26**

Summary of the Results.

- **Superior Accuracy:** Outperforms an avg. +3.43% gain.
- **Robustness:** Rescues failing optimizers (e.g., RAdam on QNLI: 49.46% → 91.78%).
- **High Efficiency:** reduces total training time by >5%.



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